Lecture 30.03: Gradient Boosting Regressor Ka Practical Implementation Aur Hyperparameters [Hinglish]











📌 Topic Overview

Practical implementation of GradientBoostingRegressor in scikit-learn. Explores synthetic regression data generation, key hyperparameters including learning_rate, n_estimators, subsample, and max_depth, along with R2 evaluation.

📋 Is Topic Mein Hum Kya Seekhenge?

- `make_regression` se synthetic continuous dataset generate karna.
- `sklearn.ensemble` se `GradientBoostingRegressor` import aur configure karna.
- Important Hyperparameters: `learning_rate`, `n_estimators`, `subsample`, aur `max_depth`.
- **Stochastic Gradient Boosting**: `subsample < 1.0` karke variance kam karna.
- Model ko R^2 score aur **MSE** ke zariye evaluate karna.

🔍 Har Ek Point Ka Simple & Detailed Explanation

- **Synthetic Data Generation**: `make_regression(n_samples=1000, n_features=10, n_informative=5, random_state=42)` se realistic regression data banaya gaya.
- **Train-Test Split**: `train_test_split(X, y, test_size=0.2, random_state=42)` se 80% training aur 20% testing data alag kiya gaya.
- **Hyperparameter 1:** `learning_rate` : Shrinkage factor (default = 0.1). Chota learning rate (0.01 - 0.1) model ko overfit hone se bachata hai.

- **Hyperparameter 2:** `n_estimators` : Kitne sequential trees banane hain (default = 100). Chote learning rate ke sath zyada trees lagte hain.
- **Hyperparameter 3:** `subsample` : **Stochastic Gradient Boosting** enable karta hai. Jaise `subsample=0.8` ka matlab har tree random 80% samples par train hoga.
- **Hyperparameter 4:** `max_depth` : Har weak tree ki depth (default = 3). Trees ko shallow rakha jata hai taaki individual trees overfit na karein.
- **Model Evaluation**: Test set par R^2 score aur `mean_squared_error` calculate karke generalization performance verify ki gayi.

📊 Visual Flowchart / Diagram



🔢 Mathematical Derivation

The mean squared error objective function:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

The coefficient of determination R^2:

$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

💻 Standalone Python Code & Verification

Iss module ka complete runnable code Jupyter Notebook mein available hai.